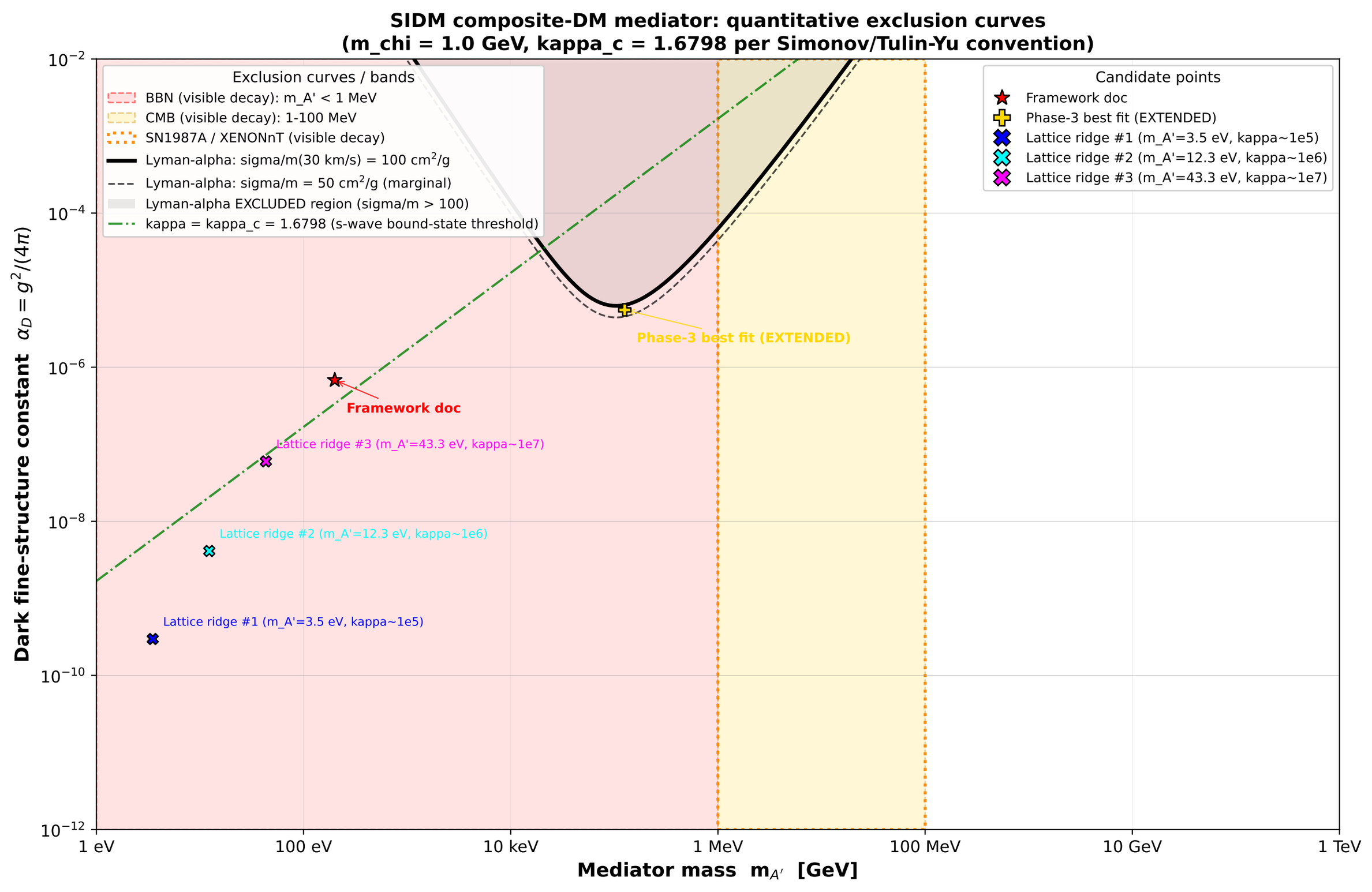

SIDM composite-DM mediator: quantitative exclusion curves
(m_chi = 1.0 GeV, kappa_c = 1.6798 per Simonov/Tulin-Yu convention)
Exclusion curves / bands
BBN (visible decay): m_A' < 1 MeV
CMB (visible decay): 1-100 MeV
SN1987A / XENONnT (visible decay)
Lyman-alpha: sigma/m(30 km/s) = 100 cm²/g
Lyman-alpha: sigma/m = 50 cm²/g (marginal)
Lyman-alpha EXCLUDED region (sigma/m > 100)
kappa = kappa_c = 1.6798 (s-wave bound-state threshold)
Candidate points
Framework doc
Phase-3 best fit (EXTENDED)
Lattice ridge #1 (m_A'=3.5 eV, kappa~1e5)
Lattice ridge #2 (m_A'=12.3 eV, kappa~1e6)
Lattice ridge #3 (m_A'=43.3 eV, kappa~1e7)
Phase-3 best fit (EXTENDED)
Framework doc
Lattice ridge #3 (m_A'=43.3 eV, kappa~1e7)
Lattice ridge #2 (m_A'=12.3 eV, kappa~1e6)
Lattice ridge #1 (m_A'=3.5 eV, kappa~1e5)
Dark fine-structure constant $\alpha_D = g^2/(4\pi)$
10^{-2}
10^{-4}
10^{-6}
10^{-8}
10^{-10}
10^{-12}
1 eV
100 eV
10 keV
1 MeV
100 MeV
10 GeV
1 TeV
Mediator mass $m_{A'}$ [GeV]